

Task 1: SQL-Based Traffic Analysis

1.1 Load the dataset

Verified that data loaded correctly.

Task 1.2 Analyse Annual Traffic trends

- Every 2 alternate years the traffic volume decreases and then increases.
- The apparent dip in data is not the actual traffic trends, as it covers few months in 2012 and 2018.

Task 1.3 Analyze temperature around holidays

Temperature during Labour Day was better and so traffic was heavy.

Task 2: Descriptive Statistics and Correlation

2.1 Traffic volume statistics

Interpret what these statistics tell you about traffic volume and its variability.

- Mean (3,259.82) is slightly below median (3,380.00), a mild left skew.
- Standard deviation (1,986.84) is large relative to the mean.
- Range (7,280) confirms a widespread between the quietest and busiest hours.

2.2 Correlation Analysis

Correlation	0.130298798
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- A correlation of 0.13 between temperature and traffic volume indicates a very weak positive relationship

Direction: The correlation coefficient is positive (0.13), meaning temperature and traffic volume move in the same direction as temperature rises, traffic volume tends to rise slightly too

Strength: At 0.13 correlation, this is a very weak relationship, close to zero.

Why does correlation not necessarily imply causation: Correlation only measures whether two variables tend to move together statistically, it says nothing about whether one is actually causing the other to change.

Task 3: Probability and Congestion Analysis

Conclude with a short explanation of what the probability analysis suggests about the relationship between weather and congestion.

Probability of congestion is 14%

Probability of clear weather is 27%

Congestion and clear weather are not independent, they're negatively associated. Clear weather makes congestion somewhat less likely. The actual co-occurrence rate is 10.6% smaller than it would be if weather and congestion had no relationship to each other.

Congestion is more likely under cloudy weather conditions. Odds ratio of congestion in clear versus cloudy weather is 0.73 which is < 1 .

Task 4: Power BI Traffic Intelligence Dashboard

4.1 Data quality and preparation

Data prepared as per the steps given. Attached a portion of the data.

task4 • Last saved: Today at 2:25 PM

Search

File Home Help **Table tools**

Name Metro_Interstate_Tr...

Manage relationships Structure

New measure Quick measure New column New table Calendar options

holiday	temperature	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume	temperature_celsius	Traffic_Category
	279.78	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3364	6.63	Low
	278.16	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3160	5.01	Low
	287.36	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2775	14.21	Low
	281.95	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3039	8.8	Low
	287.04	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2777	13.89	Low
	281.26	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2908	8.11	Low
	287.21	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2963	14.06	Low
	276.23	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3249	3.08	Low
	275.6	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3078	2.45	Low
	278.5	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2834	5.35	Low
	275.93	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2777	2.78	Low
	277.2	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2446	4.05	Low
	277.08	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2594	3.93	Low
	280.18	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2905	7.03	Low
	291.57	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2173	18.42	Low
	275.98	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3052	2.83	Low
	278.12	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	3258	4.97	Low
	284.17	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2702	11.02	Low
	277.38	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2787	4.23	Low
	265.61	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2779	-7.54	Low
	271.02	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2809	-2.13	Low
	271.29	0.0	0.0	90	Clouds	overcast clouds	8:00:00 pm	2362	-1.86	Low

4.2 Dashboard analysis

C. Weather Impact

- Which weather condition has the highest average traffic?
<Answer> shower snow
- Which has the lowest?
<Answer> thunderstorm with light rain
- What is the difference between the highest and lowest average traffic?
<Answer> 2887.17

D. Temperature and traffic

- Whether a visible relationship exists.
Traffic volume holds fairly steady in a moderate (roughly 2.8K–3.3K) across a wide middle range of temperatures, from about -25°C up through 20°C
- The temperature range associated with higher traffic.
25°C to 35°C stands out clearly as the range with the highest average traffic volume. Warm weather is when people are mostly likely to be driving
- Any notable outliers.
The point at -30°C is a striking outlier, traffic volume drops to nearly 0

4.3 KPI cards and filters

After adding slicers, depending upon the category selected, the dashboard data displays the information accordingly.